ZHIYUAN YOU

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Education

- **Ph.D. in Multimedia Lab (MMLab), The Chinese University of Hong Kong** Aug. 2023 -
  - GPA: 3.82/4.0
  - Supervisor: Professor Tianfan Xue & Professor Chao Dong
  - Direction: Generative image processing, multi-modal applications, and low-level vision
- **M.Eng. with Honor, Shanghai Jiao Tong University** Sept. 2020 - Mar. 2023
  - GPA: 3.76/4.0
  - Supervisor: Professor Xinyi Le & Professor Yu Zheng
- **B.Eng. with Honor, Shanghai Jiao Tong University** Sept. 2016 - Jun. 2020
  - GPA: 89.36/100, Ranking: 5/148
  - Supervisor: Professor Xinyi Le
  - Summer Exchange to *Columbia University* in 2018

Experience

- **NextCam Team, Adobe, Research Scientist Intern** Jul. 2025 - Nov. 2025
  - Direction: Development of VAE model with large down-sample rate and high dimensional channel.
    - ◊ Develop a detail alignment method to increase the down-sample rate and channel dimension of an existing VAE model.
    - ◊ Develop a zero-initialization method to finetune the pre-trained generation model with our new VAE model, resulting a new generation model with the same token numbers (*i.e.*, computation) and higher resolution.
  - Direction: Image composition guidance based on understanding-generation unified model.
    - ◊ Train a composition assessment model using GRPO method.
    - ◊ Finetune a composition degradation model that could degrade the good composition.
    - ◊ Construct a large-scale composition guidance dataset using various tool models (*e.g.*, composition assessment model, degradation model, image restoration model, and general MLLM) and base datasets (*e.g.*, cropping datasets, 3D scene datasets, and photography datasets).
    - ◊ Finetune a understanding-generation unified model as an image composition guidance model.
  - Mentor: Dr. Zhoutong Zhang, Dr. Ke Wang, and Dr. He Zhang
- **Shanghai Artificial Intelligence Laboratory, Research Intern** May 2024 - Jan. 2025
  - Mentor: Professor Chao Dong

- **Shenzhen Institutes of Advanced Technology, Research Intern** Apr. 2023 - Jul. 2023
  - Mentor: Professor Chao Dong
- **Horizon Robotics, Research Intern** Nov. 2022 - Mar. 2023
  - Mentor: Dr. Yuelong Yu
- **SenseTime, Research Intern** Dec. 2020 - Nov. 2022
  - Mentor: Mr. Kai Yang

Publications

- M. Yao, **Z. You**, T. Man, M. Wang, T. Xue, "PhotoAgent: Agentic Photo Editing with Exploratory Visual Aesthetic Planning," in Proc. of International Conference on Machine Learning (**ICML, Spotlight**), 2026.
- J. Wang, H. Wu, **Z. You**, Y. Song, Y. Wang, Z. Shan, Y. Li, S. Zhang, X. Le, C. Chen, X. Guan, D. Tao, "RouteMoA: Dynamic Routing without Pre-Inference Boosts Efficient Mixture-of-Agents," in Proc. of Annual Meeting of the Association for Computational Linguistics (**ACL**), 2026.
- X. Cai, **Z. You**, Z. Zhang, T. Xue, "DA-VAE: Plug-in Latent Compression for Diffusion via Detail Alignment," in Proc. of IEEE/CVF Conference on Computer Vision and Pattern Recognition (**CVPR**), 2026.
- **Z. You**, K. Wang, H. Zhang, X. Cai, J. Gu, T. Xue, C. Dong, Z. Zhang, "PhotoFramer: Multi-modal Image Composition Instruction," in Proc. of IEEE/CVF Conference on Computer Vision and Pattern Recognition (**CVPR**), 2026.
- X. Yin, J. Hu, **Z. You**, K. Yan, Y. Tang, C. Dong, J. Gu, "How Far Have We Gone in Generative Image Restoration? A Study on Its Capability, Limitations and Evaluation Practices," in Proc. of IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshop (**CVPRW**), 2026.
- X. He*, **Z. You***, J. Gong, C. Liu, X. Yue, P. Zhuang, W. Zhang, L. Bai, "RadarQA: Multi-modal Quality Analysis of Weather Radar Forecasts," in Proc. of Annual Conference on Neural Information Processing Systems (**NeurIPS**), 2025.
- X. Wang, J. Zhuo, **Z. You**, Z. Tan, Y. Yu, S. Wang, X. Le, "ReinAD: Towards Real-world Industrial Anomaly Detection with a Comprehensive Contrastive Dataset," in Proc. of Annual Conference on Neural Information Processing Systems (**NeurIPS**), Datasets and Benchmarks Track, 2025.
- X. Lin, F. Yu, J. Hu, **Z. You**, W. Shi, J. S. Ren, J. Gu, C. Dong, "Harnessing Diffusion-Yielded Score Priors for Image Restoration," in ACM **SIGGRAPH Asia**, ACM Transactions on Graphics (**TOG**), 2025.
- J. Gao, R. Liu, Y. Peng, S. Yang, J. Zhang, K. Yang, **Z. You**, "DeQA-Doc: Adapting DeQA-Score to Document Image Quality Assessment," in Proc. of IEEE/CVF International Conference on Computer Vision Workshop (**ICCVW**), 2025.
- J. Hu*, **Z. You***, J. Gu, K. Zhu, T. Xue, C. Dong, "Revisiting the Generalization Problem of Low-level Vision Models Through the Lens of Image Deraining," Pattern Recognition (**PR**), 2025.
- J. Hu, J. Gu, S. Yu, F. Yu, Z. Li, **Z. You**, C. Lu, C. Dong, "Interpreting Low-level Vision Models with Causal Effect Maps," IEEE Transactions on Pattern Analysis and Machine Intelligence (**TPAMI**), 2025.
- **Z. You**, X. Cai, J. Gu, T. Xue, C. Dong, "Teaching Large Language Models to Regress Accurate Image Quality Scores using Score Distribution," in Proc. of IEEE/CVF Conference on Computer Vision and Pattern Recognition (**CVPR**), 2025.
- Z. Chen, Y. Wang, X. Cai, **Z. You**, Z. Lu, F. Zhang, S. Guo, T. Xue, "UltraFusion: Ultra High Dynamic Imaging using Exposure Fusion," in Proc. of IEEE/CVF Conference on Computer Vision and Pattern Recognition (**CVPR, Highlight, Best Demo Honorable Mention**), 2025.

- K. Zhu, J. Gu, **Z. You**, Y. Qiao, C. Dong, "An Intelligent Agentic System for Complex Image Restoration Problems," in Proc. of International Conference on Learning Representations (**ICLR**), 2025.
- J. Zhang*, **Z. You***, J. Wang, X. Le, "SAIL: Sample-Centric In-Context Learning for Document Information Extraction," in Proc. of Association for the Advancement of Artificial Intelligence (**AAAI, Oral**), 2025.
- **Z. You**, X. Cai, J. Gu, Z. Li, K. Zhu, C. Dong, T. Xue, "Enhancing Descriptive Image Quality Assessment with A Large-scale Multi-modal Dataset," IEEE Transactions on Image Processing (**TIP**), 2025
- X. Cai, **Z. You**, H. Zhang, W. Liu, J. Gu, T. Xue, "PhoCoLens: Photorealistic and Consistent Reconstruction in Lensless Imaging," in Proc. of Annual Conference on Neural Information Processing Systems (**NeurIPS, Spotlight**), 2024.
- **Z. You**, Z. Li, J. Gu, Z. Yin, T. Xue, C. Dong, "Depicting Beyond Scores: Advancing Image Quality Assessment through Multi-modal Language Models," in Proc. of European Conference on Computer Vision (**ECCV**), 2024.
- J. Liu, Y. Zhang, C. Li, **Z. You**, Z. Zhou, C. Yang, Y. Yang, Y. Liu, W. Ouyang, "MaskMA: Towards Zero-Shot Multi-Agent Decision Making with Mask-Based Collaborative Learning," Transactions on Machine Learning Research (**TMLR**), 2024.
- **Z. You**, K. Yang, W. Luo, X. Lu, L. Cui, X. Le, "Few-Shot Object Counting with Similarity-Aware Feature Enhancement," in Proc. of IEEE Winter Conference on Applications of Computer Vision (**WACV, Early Accept**), 2023.
- **Z. You**, L. Cui, Y. Shen, K. Yang, X. Lu, Y. Zheng, X. Le, "A Unified Model for Multi-class Anomaly Detection," in Proc. of Annual Conference on Neural Information Processing Systems (**NeurIPS, Spotlight**), 2022.
- **Z. You**, K. Yang, W. Luo, L. Cui, Y. Zheng, X. Le, "ADTR: Anomaly Detection Transformer with Feature Reconstruction," in Proc. of International Conference on Neural Information Processing (**ICONIP, Oral**), 2022.
- **Z. You**, J. Li, H. Zhang, B. Yang, X. Le, "An Accurate Star Identification Approach Based on Spectral Graph Matching for Attitude Measurement of Spacecraft," Complex & Intelligent Systems, 2022.
- L. Chen, **Z. You**, N. Zhang, J. Xi, X. Le, "UTRAD: Anomaly Detection and Localization with U-Transformer," Neural Networks, 2022.

Services

- **Journal Reviewer**
  - IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI)
  - International Journal of Computer Vision (IJCV)
  - IEEE Transactions on Image Processing (TIP)
  - IEEE Transactions on Multimedia (TMM)
  - IEEE Transactions on Neural Networks and Learning Systems (TNNLS)
  - IEEE Transactions on Systems, Man and Cybernetics: Systems (TSMCS)
  - IEEE Transactions on Circuits and Systems for Video Technology (TCSVT)
  - IEEE Transactions on Knowledge and Data Engineering (TKDE)
  - Pattern Recognition (PR)
  - Knowledge Based Systems (KBS)
  - Engineering Applications of Artificial Intelligence (EAAI)
  - Neural Networks (NN)

- Journal of Intelligent Manufacturing
- Journal of Selected Topics in Signal Processing
- Neurocomputing
- Measurement

- **Conference Reviewer**
  - IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2024, 2025, 2026
  - International Conference on Computer Vision (ICCV), 2025
  - Annual Conference on Neural Information Processing Systems (NeurIPS), 2024, 2025
  - International Conference on Learning Representations (ICLR), 2025, 2026
  - International Conference on Machine Learning (ICML), 2025
  - Association for the Advancement of Artificial Intelligence (AAAI), 2026
  - ACM International Conference on Multimedia (MM), 2026
  - International Conference on Artificial Intelligence and Statistics (AISTATS), 2025, 2026
  - The British Machine Vision Conference (BMVC), 2026
  - IEEE Winter Conference on Applications of Computer Vision (WACV), 2026

Honors and Awards

- CUHK Reaching Out Award 2025
- SJTU Excellent Master Dissertation 2023
- SJTU Outstanding Graduate (Postgraduate) 2023
- National Scholarship 2022
- SJTU Outstanding Graduate (Undergraduate) 2020
- SJTU ABB Scholarship 2017, 2018, 2019
- SJTU Tingya Scholarship 2019
- National Encourage Scholarship 2017

Teaching

- **Teaching Assistant, The Chinese University of Hong Kong** Jan. 2025 - May. 2025
  - Course: IERG4210, Web Programming and Security.
- **Teaching Assistant, The Chinese University of Hong Kong** Sept. 2024 - Dec. 2024
  - Course: IERG4230, Introduction to Internet of Things.
- **Teaching Assistant, The Chinese University of Hong Kong** Sept. 2024 - Dec. 2024
  - Course: IERG2051, Signals and Systems.
- **Teaching Assistant, The Chinese University of Hong Kong** Jan. 2024 - May. 2024
  - Course: IERG4160, Image and Video Processing.
- **Teaching Assistant, The Chinese University of Hong Kong** Sept. 2023 - Dec. 2023
  - Course: ENGG2720, Complex Variables for Engineers.
- **Teaching Assistant, Shanghai Jiao Tong University** Sept. 2020 - Jan. 2021
  - Course: ME28009, Dynamic Programming and Optimal Control.